ALKENE

141. In the reaction

$$CH_3CH = CH_2 + H_2O + [O] \xrightarrow[Acid]{KMnO_4}$$

$$CH_3 - \underset{\underset{OH}{|}}{CH} - \underset{\underset{OH}{|}}{CH_2} \xrightarrow{[O]} X + HCOOH$$

X is

(a) CH_3CH_2COOH

(b) CH_3COOH

(c) CH_3CH_2CHO

(d) CH_3CH_2OH

142. Which of the following alkenes gives only acetic acid and on oxidation with potassium permanganate solution

(a) Ethylene (b) 1-Butene

(c) Propene (d) 2-Butene

143. Butene-1 may be converted to butane by reaction with

(a) *Zn-HCl* (b) *Sn-HCl*

(c) *Zn-Hg* (d) *Pd*/H_2

144. The major product formed when propene reacts with *HBr* in presence of peroxides is

(a) *n*-propyl bromide

(b) Isopropyl bromide

(c) *n*-propyl alcohol

(d) 1, 3-dibromopropane

145. Ethyl hydrogen sulphate is obtained by the reaction of H_2SO_4 on

(a) Ethylene (b) Ethane

(c) Ethyl chloride (d) Ethanol

146. Ethylene reacts with ozone to give

(a) Formaldehyde

(b) Ethyl alcohol

(c) Ozonide

(d) Acetaldehyde

147. Which of the following aliphatic compounds will discharge red colour of bromine

(a) C_2H_4 (b) C_3H_6

(c) C_4H_8 (d) All of these

148. Chlorination can be done on

(a) $CH_3 - CH = Ch_2$

(b) $CH_2 = CH_2$

(c) $CH \equiv CH$

(d) None of these

149. Addition of *HI* on the double bond of propene yields isopropyl iodide and not *n*-propyl iodide as the major product. This is because the addition proceeds through

(a) A more stable carbonium ion

(b) A more stable carbanion

(c) A more stable free radical

(d) None of the above being a concerted reaction

150. When butene-1 is mixed with excess of bromine, the expected reaction product is

(a) 1, 2-dibromobutane

(b) 1, 1-dibromobutane

(c) 2, 2-dibromobutane

(d) Perbromobutane

151. A compound '*X*' on ozonolysis forms two molecules of *HCHO*. Compound '*X*' is

(a) C_2H_4 (b) C_2H_2

(c) C_2H_6 (d) C_6H_6

152. For the reaction

$CH_3 - CH = CH_2 + HOCl \rightarrow A$ the product *A* is

(a) $CH_3 - CHCl - CH_2OH$

(b) $CH_3 - \underset{\underset{OH}{|}}{CH} - CH_2 - Cl$

(c) $CH_3 - CH_2 - CH_2 - COCl$

(d) $CH_3 - \overset{\overset{Cl}{|}}{\underset{\underset{OH}{|}}{C}} - CH_3$

153. $(CH_3)_2C = \underset{\underset{CH_3}{|}}{CH} \xrightarrow[H_2]{\text{Catalyst}}$ Optical isomers

(a) 2 (b) 4

(c) Zero (d) 3

154. Isobutene $+HBr \xrightarrow[\text{Peroxide}]{}$ product is

(a) Tertiary butyl bromide

(b) Isobutyl bromide

(c) Tertiary butyl alcohol

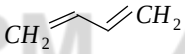
(d) Isobutyl alcohol

155. Which of the following represents the given mode of hybridisation sp^2–sp^2– *sp*-*sp* from left to right

(a) $H_2C = CH - C \equiv CH$

(b) $HC \equiv C - C \equiv CH$

(c) $H_2C = C = C = CH_2$

(d) 

156. "The negative part of addenda adds on to the carbon atom linked with least number of hydrogen atoms". This statement is called

(a) Thiele's principle

(b) Bayer's strain theory

(c) Markownikoff's rule

(d) Peroxide effect

157. The product obtained, heating ethanol with conc. H_2SO_4 at $165^o - 170^o$, is

(a) $(C_2H_5)_2SO_4$ (b) $CH_2 = CH_2$

(c) CH_3COOH (d) $C_2H_5HSO_4$

158. Which of the following is the most stable

(a) 1-butene (b) 2-butene

(c) 1-pentene (d) 2-pentene

159. Which doesn't follow Markownikoff's rule

(a) $CH_3 - CH = CH_2$

(b) $CH_3CH = CHCH_3$

(c) $CH_3 - \underset{\underset{CH_3}{|}}{CH} - CH = CH_2$

(d) $\boldsymbol{CH_3 - CH_2 - CH = CH_2}$

160. The product of acid catalyzed hydration of 2-phenyl propene is

(a) 3-phenyl-2-propanol

(b) 1-phenyl-2-propanol

(c) 2-phenyl-2-propanol

(d) 2-phenyl-1-propanol